MapReduce is not the only option available within LangChain. The framework also supports other summarization strategies through the `loadSummarizationChain` function, including the stuffing and refine methods.

The *stuffing* method involves passing the entire document to the language model in a single call. It "stuffs" the text into the prompt as context, allowing the model to generate a summary based on all available information at once. This is the preferred choice for shorter documents that fit inside the context window.

The *refine* method operates by first generating an initial summary from a small chunk of data. Subsequent chunks are then processed, with this initial output being refined based on new information. This is the preferred choice for larger documents where context from previous outputs is used to improve the final output. Figure 7.3 demonstrates how the refine summarization process works.

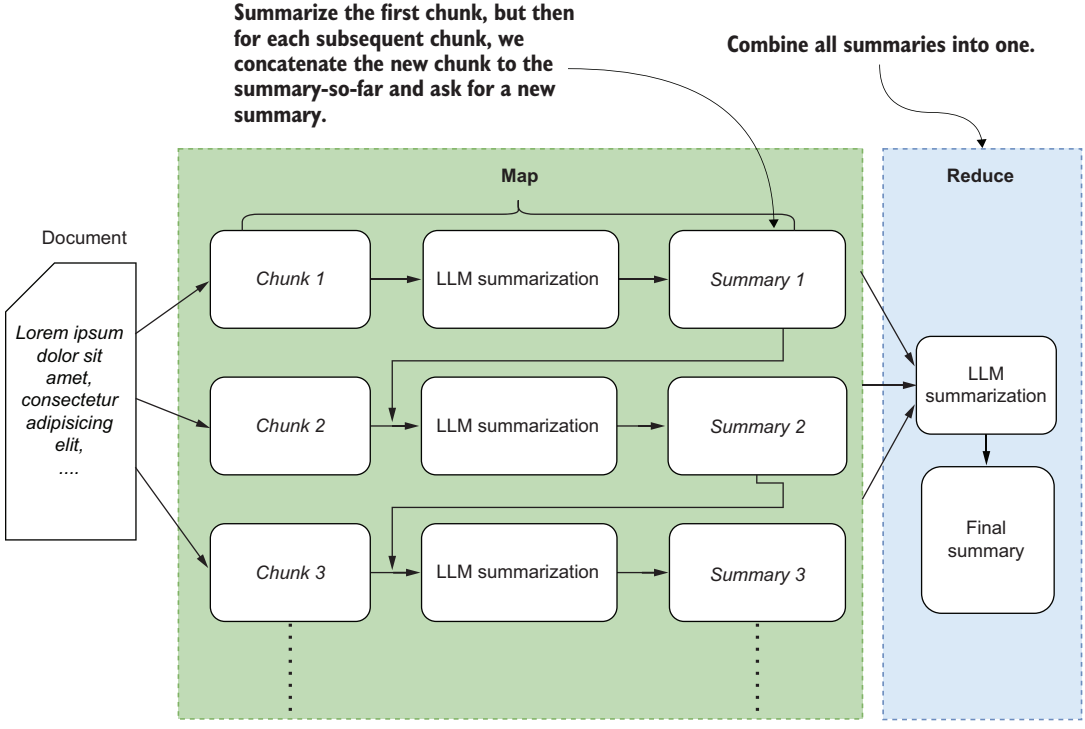


Figure 7.3 How the refine summarization method operates behind the scenes. This method enhances summarization quality by iteratively updating the summary with each new chunk of text. Initially, the first chunk is summarized, and for each subsequent chunk, it is concatenated to the existing summary. A new summary is generated based on this updated text, repeating the process until all chunks have been processed.

Each of these two methods has its advantages and tradeoffs, which you will need to consider when you expose any application that uses them to the public. For example, the